



Chapter 19

Network Layer: Logical Addressing

Logical Addressing

- *Communication at network layer is host-to-host.*
- *Packets sent may pass through several LANs & WANs.*
- *For this level of communication a global addressing scheme is needed called **Logical addressing**.*
- *Internet addresses are 32 bits in length gives a maximum of 2^{32} addresses. These addresses are called IPv4 addresses or simply IP addresses.*
- *For need of more addresses internet uses 128-bit addresses that give greater flexibility in address allocation. These addresses are called IPv6 addresses*

19-1 IPv4 ADDRESSES

*An **IPv4 address** is a **32-bit** address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.*

The IPv4 addresses are unique and universal.

- *IP addresses are unique in the sense that each address defines only one connection to the internet.*
- *Two devices on the internet can never have the same IP at the same time.*
- *If a device operating at network layer has m connections to the internet it need to have m addresses. **Eg. Router***
- *IP addresses are unique in the sense that the addressing system must be accepted by any host that want to connected to internet.*

Address Space

**The address space of IPv4 is
 2^{32} or 4,294,967,296.**

- *An address space is the total number of addresses used by the protocol.*
- *If protocol uses N bits to define an address, the address space is 2^N .*
- *The address space of IPv4 is more than four billion.*

Notations

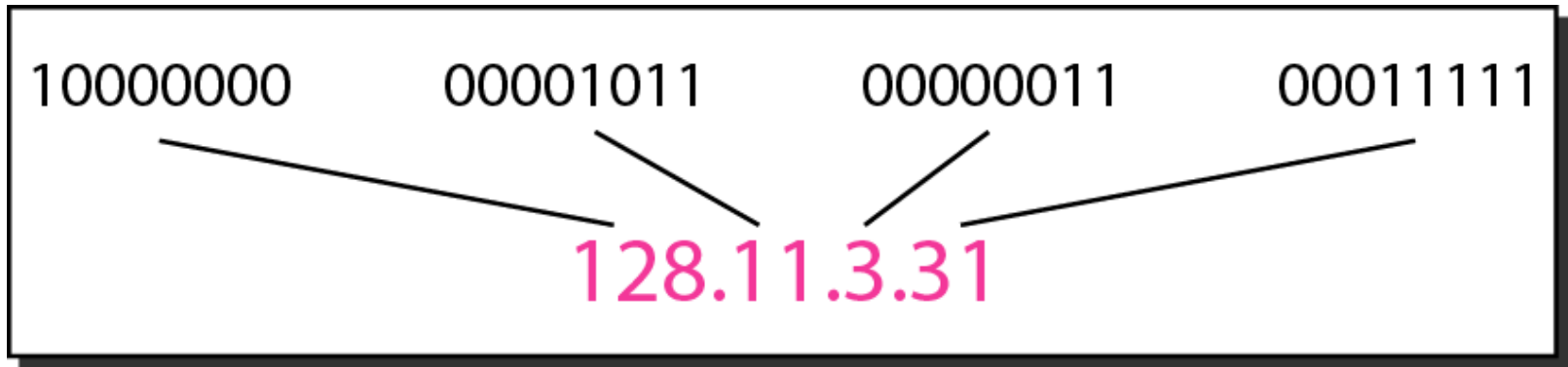
Binary Notation:

In this notation address is displayed as 32 bits.

Dotted Decimal Notation:

To make IP address more compact and easier to read this notation is used.

Each number in dotted decimal notation is a value ranging from 0 to 255.





Example 19.1

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255



Example 19.2

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010



Example 19.3

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. There must be no leading zero (045).***
- b. There can be no more than four numbers.***
- c. Each number needs to be less than or equal to 255.***
- d. A mixture of binary notation and dotted-decimal notation is not allowed.***

Classful Addressing

IPv4 uses the concept of classes. The architecture is called Classful Addressing.

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

The class of an address can be find out if address is given in either binary or decimal form.

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation



Example 19.4

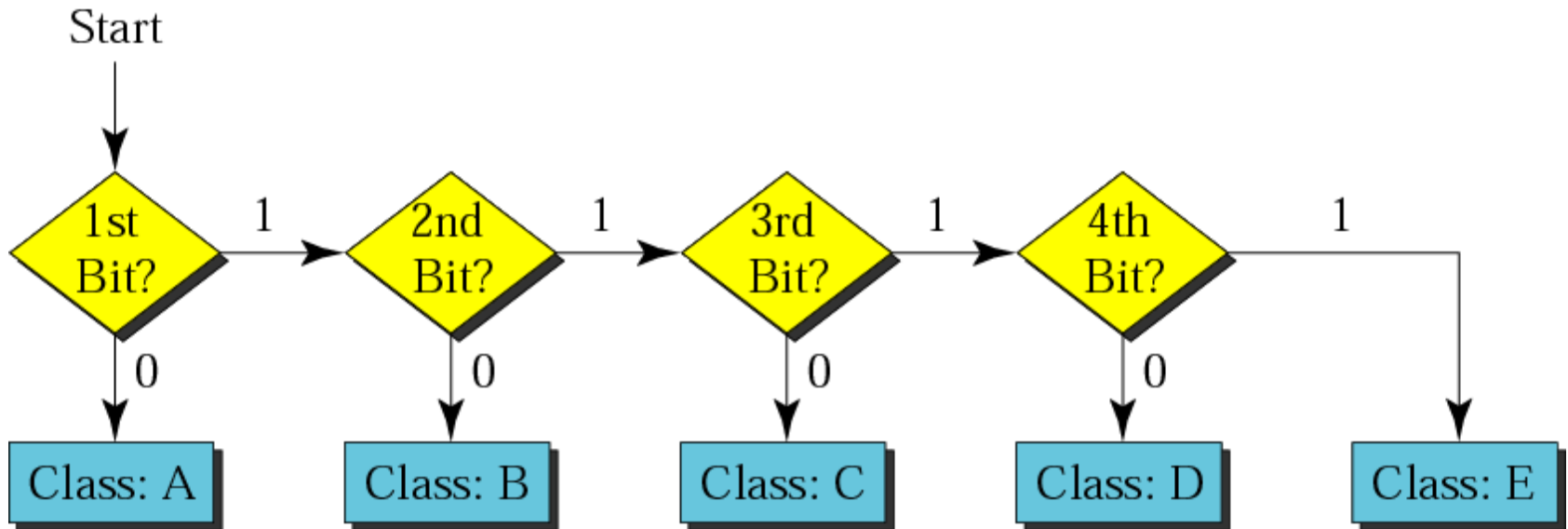
Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Finding the address class



Classes and Blocks

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Class A addresses were designed for large organizations with large number of hosts.

Class B addresses were designed for midsize organizations with tens of thousands of hosts.

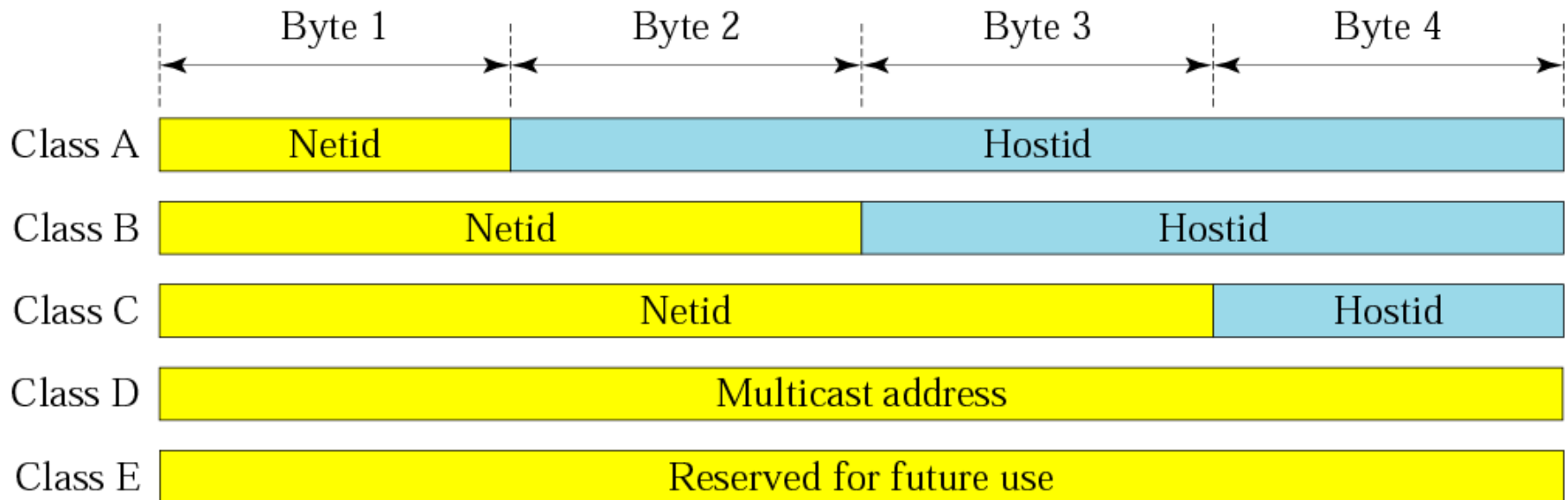
Class A addresses were designed for small organizations with small number of hosts.

Class D addresses were designed for multicasting.

19.14 Class E addresses were reserved for future use.

Netid and Hostid

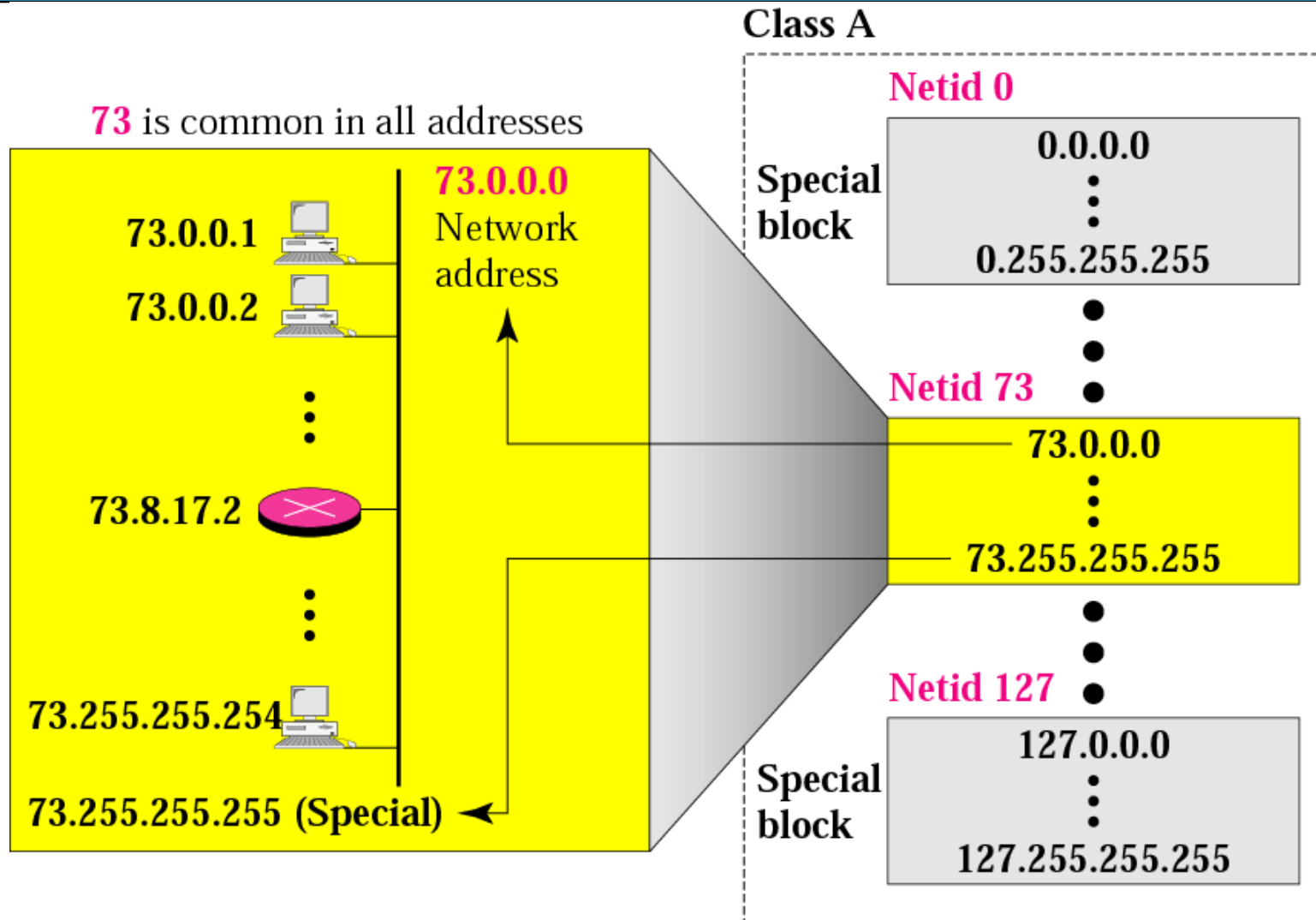
IP addresses in class A, B and C is divided into netid and hostid. This concept does not apply to class D and E.



In class A one byte defines the netid and three byte define the hostid and so on.

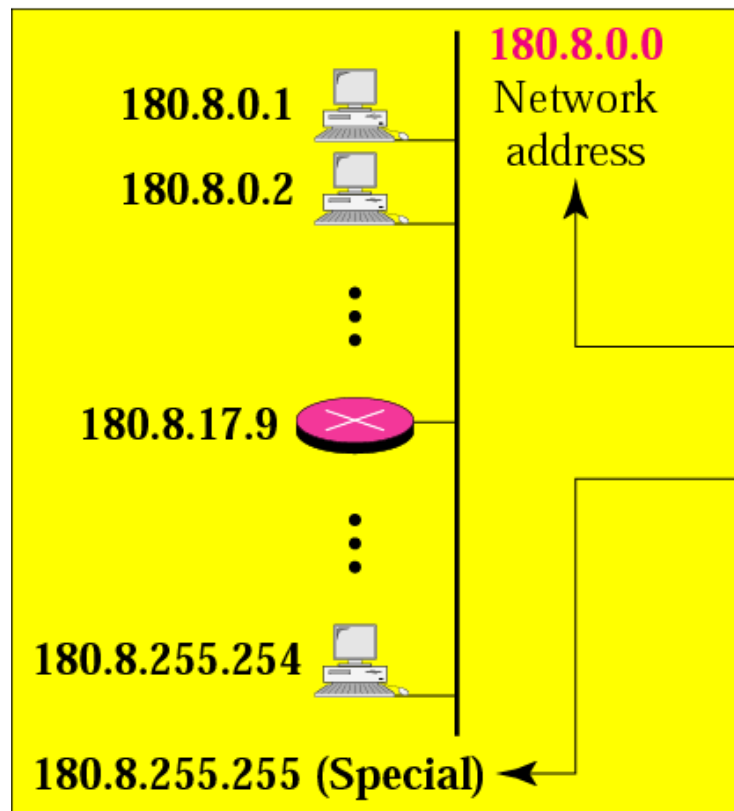
In classful addressing, a large part of the available addresses were wasted.

Blocks in class A



Blocks in class B

180.8 is common in all addresses



Class B

Netid 128.0

128.0.0.0
⋮
128.0.255.255

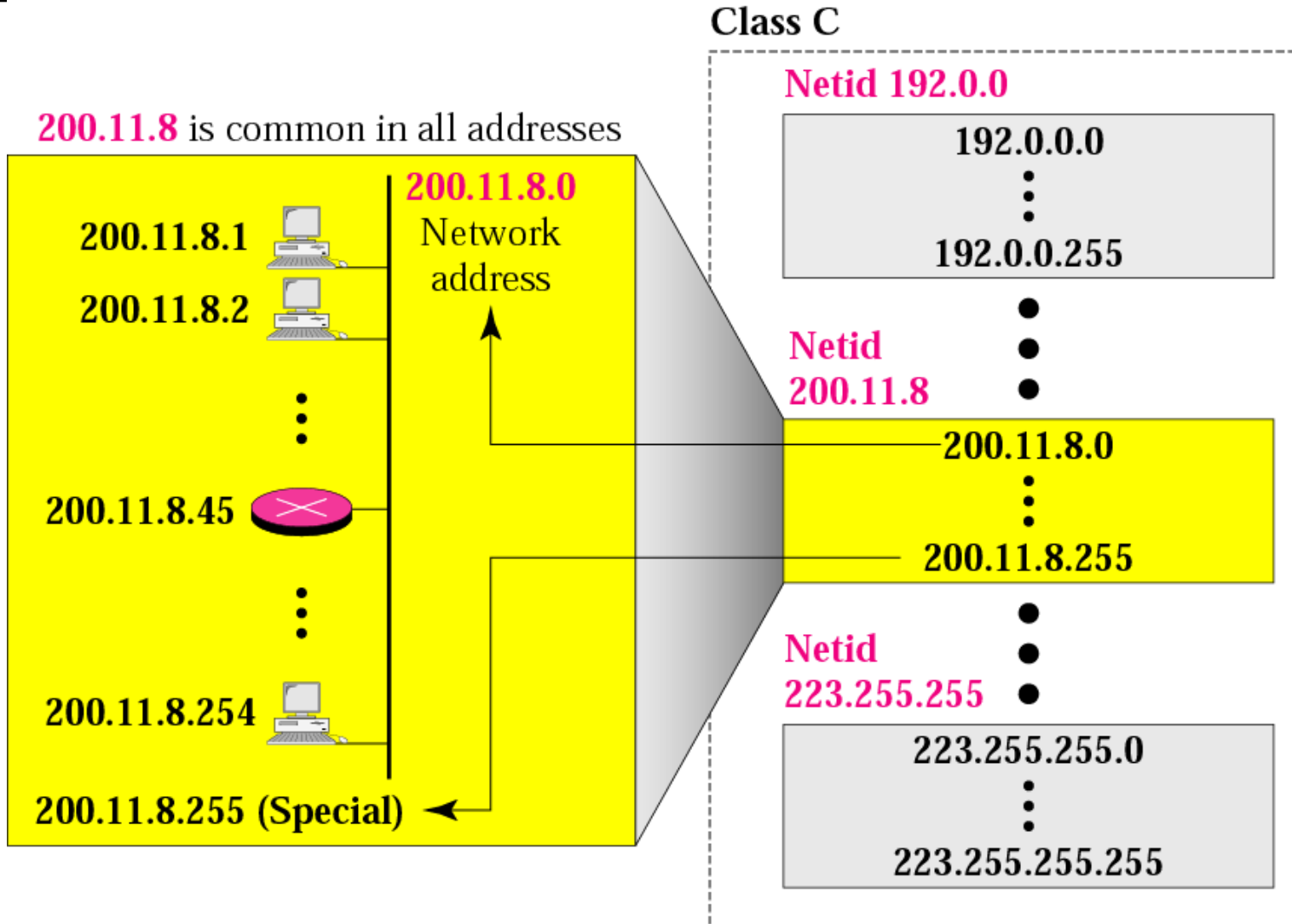
Netid 180.8

180.8.0.0
⋮
180.8.255.255

Netid 191.255

191.255.0.0
⋮
191.255.255.255

Blocks in class C



Network Addresses

- The network address is the first address.
- The network address defines the network to the rest of the Internet.
- Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

Example 9

Given the network address 17.0.0.0, find the class, the block, and the range of the addresses.

Solution

The class is A because the first byte is between 0 and 127. The block has a netid of 17. The addresses range from 17.0.0.0 to 17.255.255.255.

Example 10

Given the network address 132.21.0.0, find the class, the block, and the range of the addresses.

Solution

The class is B because the first byte is between 128 and 191. The block has a netid of 132.21. The addresses range from 132.21.0.0 to 132.21.255.255.

Example 11

Given the network address 220.34.76.0, find the class, the block, and the range of the addresses.

Solution

The class is C because the first byte is between 192 and 223. The block has a netid of 220.34.76. The addresses range from 220.34.76.0 to 220.34.76.255.

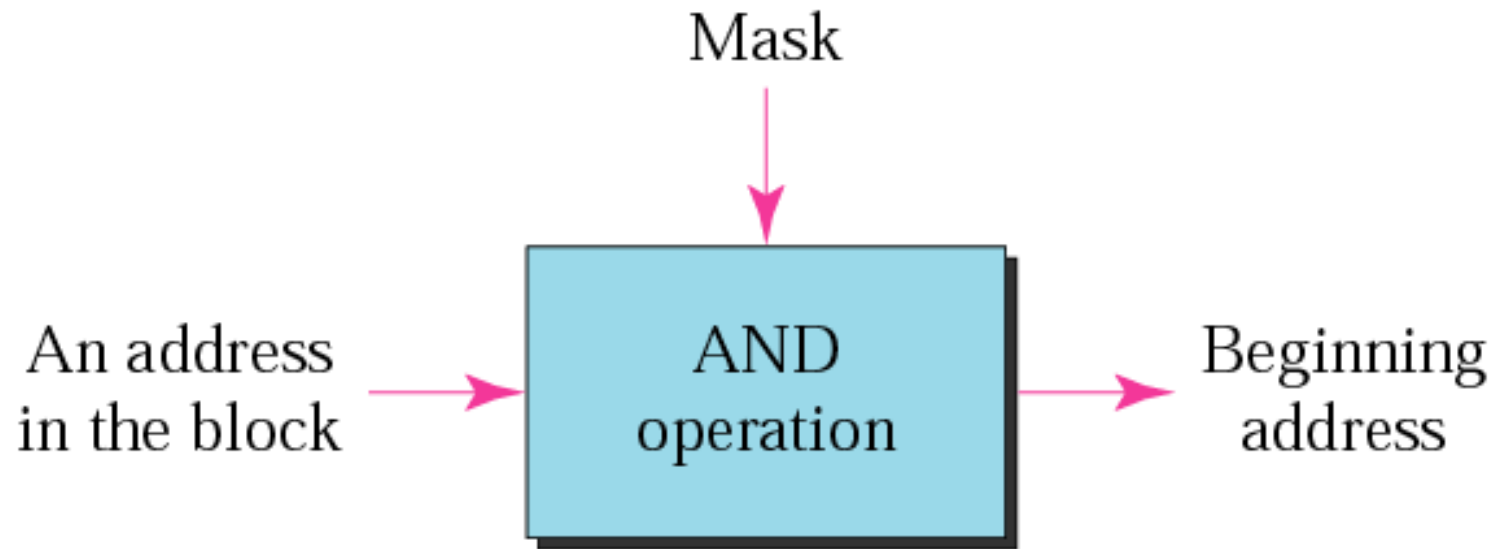
Mask

A mask is a 32-bit binary number that gives the first address in the block (the network address) when bitwise ANDed with an address in the block.

Mask

- Mask also called Default mask is a 32 bit number made of contiguous 1s followed by contiguous 0s.
- Mask helps in finding netid and hostid.
- Example:** Mask for class A address has eight 1s i.e. 8 bits of any address in class A define the netid and rest 24 bits define the hostid.
- Mask in the form /**n** (where n= 8, 16, 24 in classful addressing) is also used.
- This notation is called slash notation or **Classless Interdomain Routing(CIDR)** notation. This notation is used in classless addressing.

Masking concept



AND operation

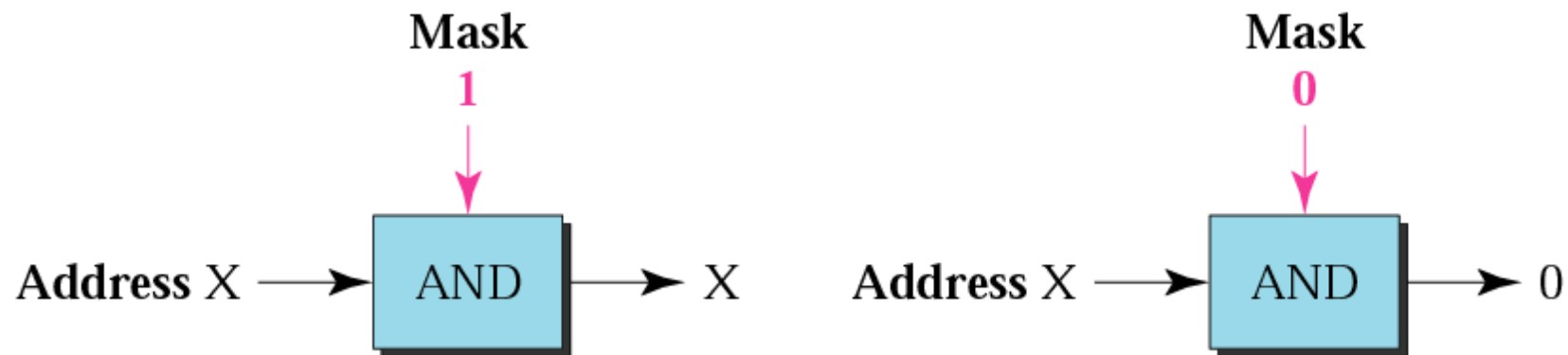


Table 19.2 *Default masks for classful addressing*

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

Note

*The network address is the beginning address of each block. It can be found by applying the default mask to any of the addresses in the block (including itself). It retains the **netid** of the block and sets the **hostid** to zero.*

Example 12

Given the address 23.56.7.91 and the default class A mask, find the beginning address (network address).

Solution

The default mask is 255.0.0.0, which means that only the first byte is preserved and the other 3 bytes are set to 0s. The network address is 23.0.0.0.

Example 13

Given the address 132.6.17.85 and the default class B mask, find the beginning address (network address).

Solution

The default mask is 255.255.0.0, which means that the first 2 bytes are preserved and the other 2 bytes are set to 0s. The network address is 132.6.0.0.

Example 14

Given the address 201.180.56.5 and the class C default mask, find the beginning address (network address).

Solution

The default mask is 255.255.255.0, which means that the first 3 bytes are preserved and the last byte is set to 0. The network address is 201.180.56.0.

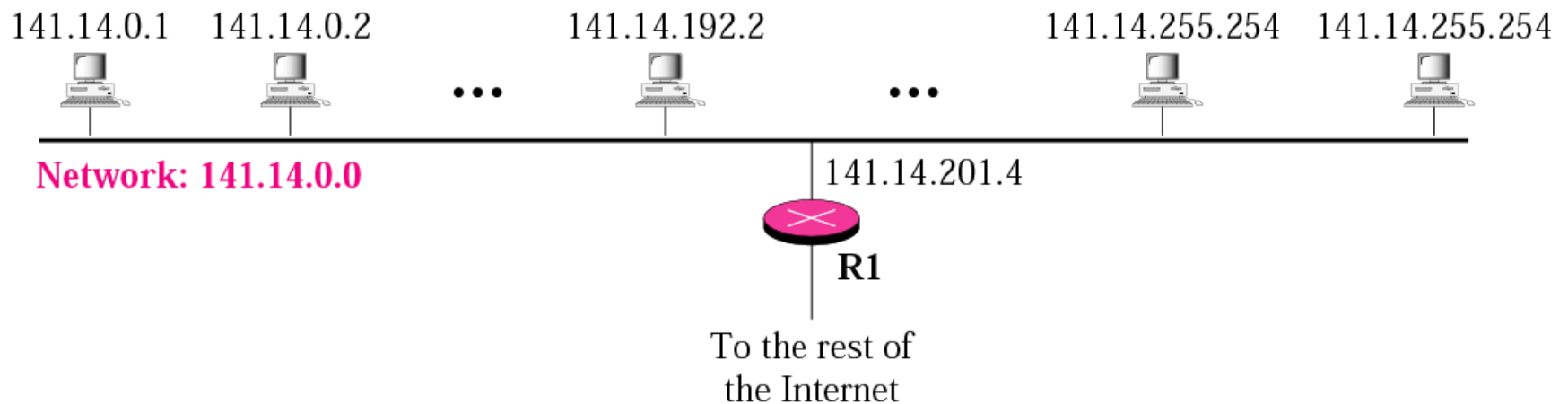
Note

*We must not
apply the default mask
of one class to
an address belonging
to another class.*

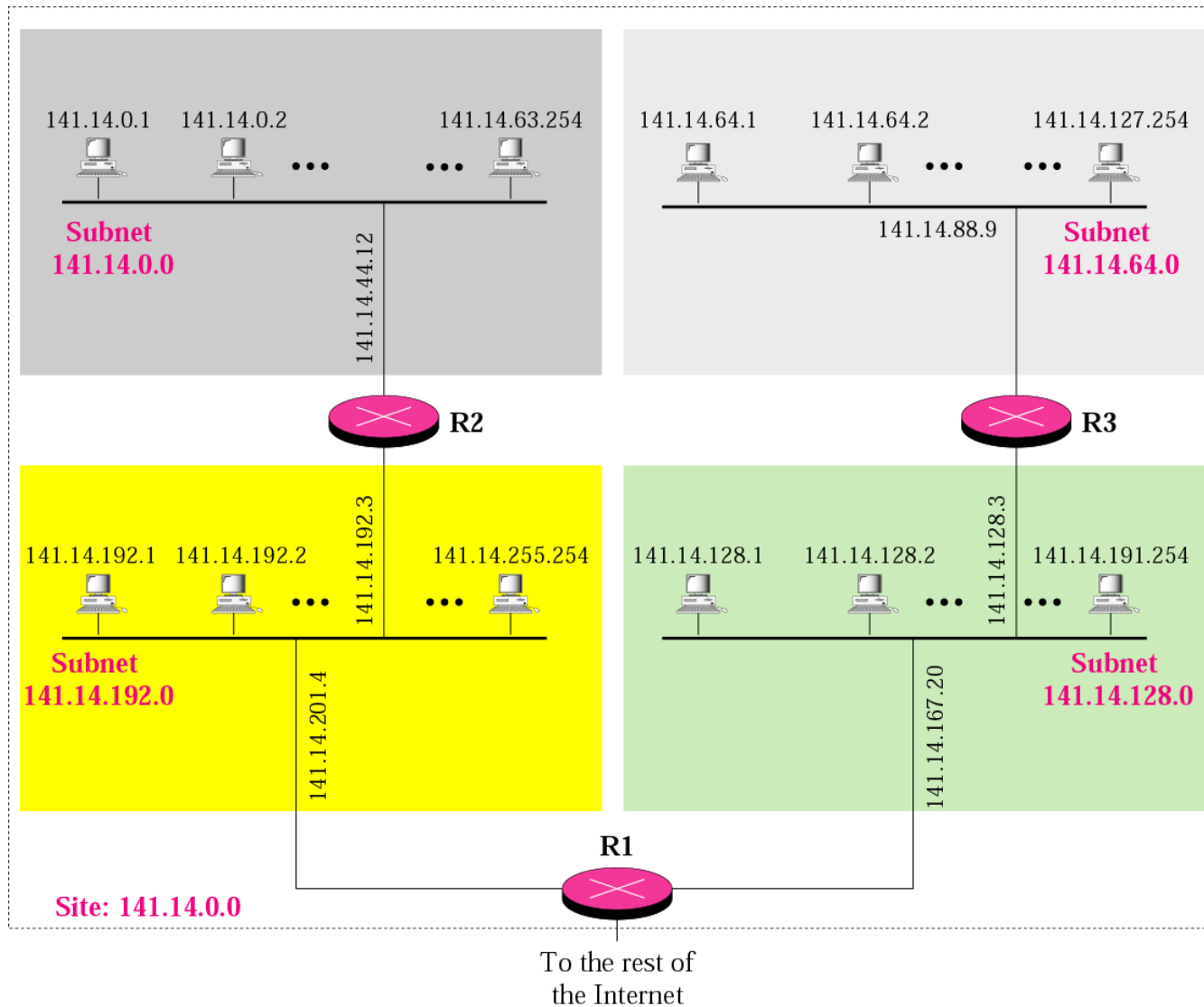
Subnetting & Supernetting

- If an organization was granted a large block in class A or B, it could divide the addresses into small groups called subnets.
- Subnetting** increases the number of 1s in mask(In classless addressing).
- In supernetting**, an organization can combine several class C blocks to create a larger range of addresses.
- Several networks are combined to create a supernetwork or supernet.
- Supernetting decreases the number of 1s in the mask(In classless addressing).

A network with two levels of hierarchy (not subnetted)

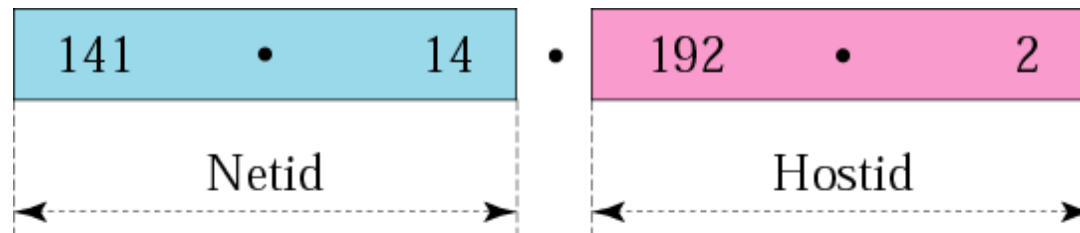


A network with three levels of hierarchy (subnetted)

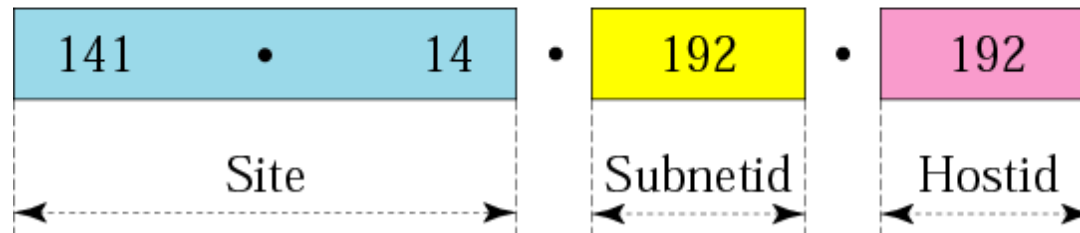


Addresses in a network with and without subnetting

For making 4 subnets 2 bits are set to 1

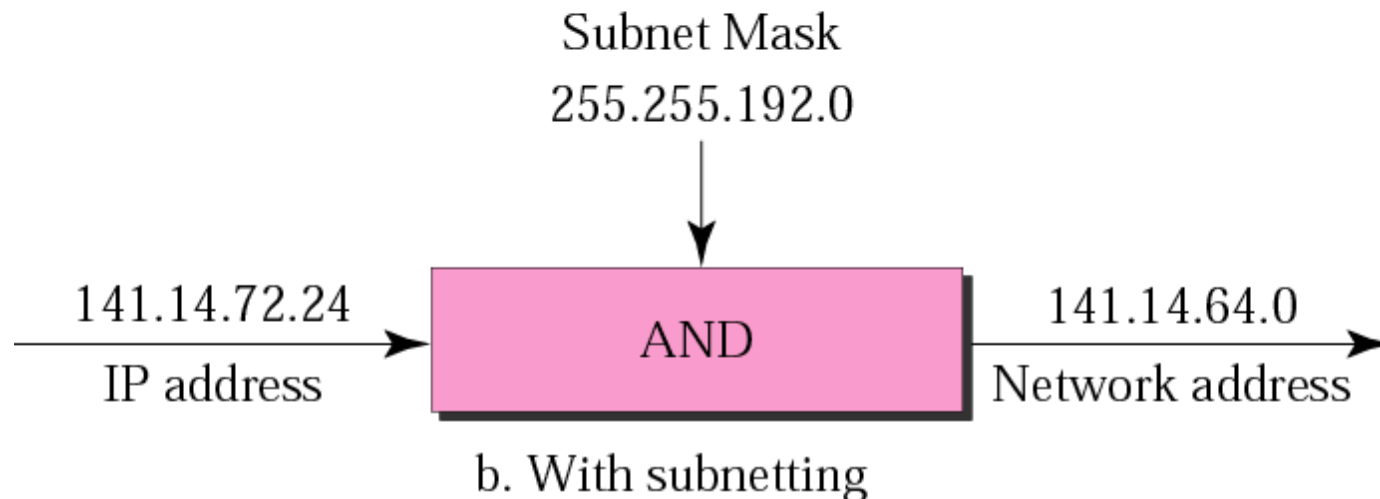
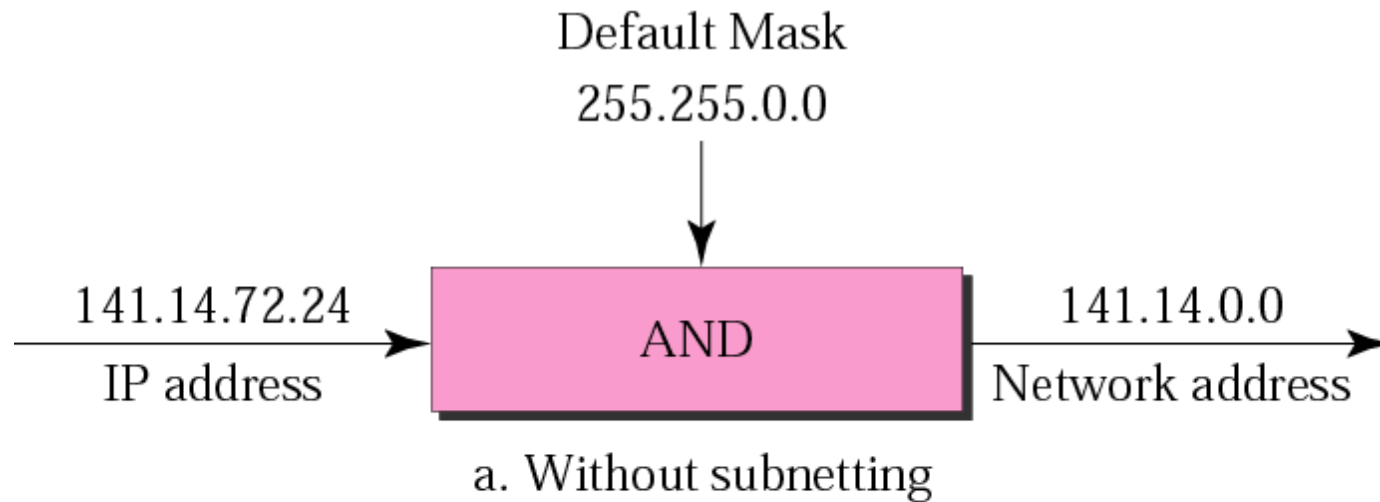


a. Without subnetting



b. With subnetting

Default mask and subnet mask



Finding the Subnet Address

Given an IP address, we can find the subnet address the same way we found the network address in the previous chapter. We apply the mask to the address. We can do this in two ways: straight or short-cut.

Straight Method

In the straight method, we use binary notation for both the address and the mask and then apply the AND operation to find the subnet address.

Example 1

What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?

Solution

11001000 00101101 00100010 00111000

11111111 11111111 11110000 00000000

11001000 00101101 00100000 00000000

The subnetwork address is 200.45.32.0.

Short-Cut Method

- ** If the byte in the mask is 255, copy the byte in the address.
- ** If the byte in the mask is 0, replace the byte in the address with 0.
- ** If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

Example 2

What is the subnetwork address if the destination address is 19.30.80.5 and the mask is 255.255.192.0?

Solution

See Figure 5.6

Example 2

IP Address

19	•	30	•	84	•	5
----	---	----	---	----	---	---

Mask

255	•	255	•	192	•	0
-----	---	-----	---	-----	---	---

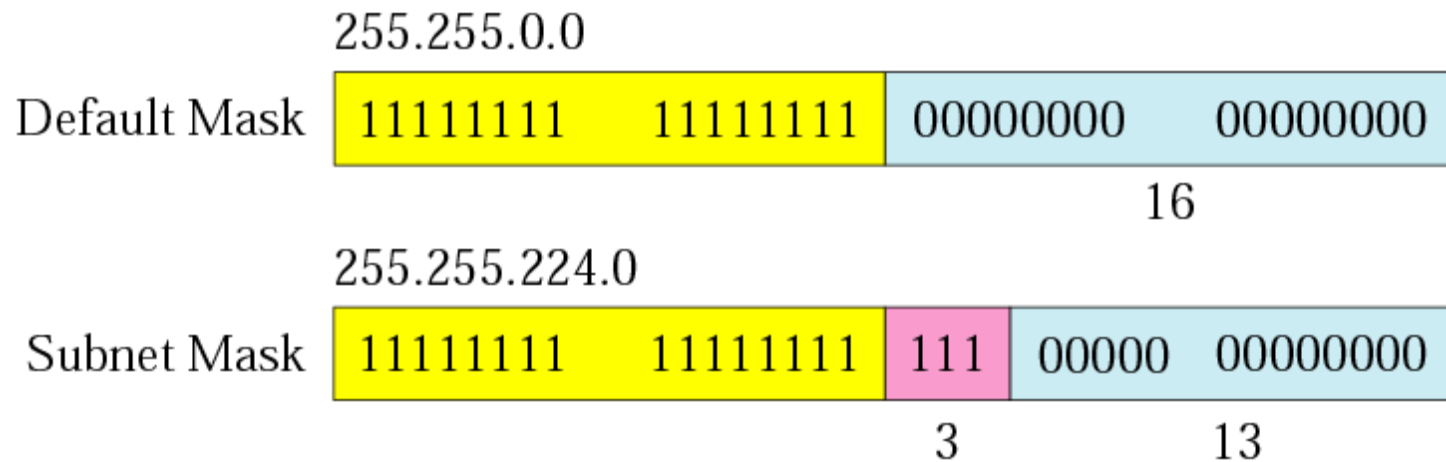
19	•	30	•	64	•	0
----	---	----	---	----	---	---

Subnet Address

↓

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0

Comparison of a default mask and a subnet mask



Note

The number of subnets must be a power of 2.

Example 3

A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.

Solution

The number of 1s in the default mask is 24 (class C).

Solution (Continued)

The company needs six subnets. This number 6 is not a power of 2. The next number that is a power of 2 is 8 (2^3). We need 3 more 1s in the subnet mask. The total number of 1s in the subnet mask is 27 ($24 + 3$).

The total number of 0s is 5 ($32 - 27$). The mask is

Solution (Continued)

11111111 11111111 11111111 11100000

or

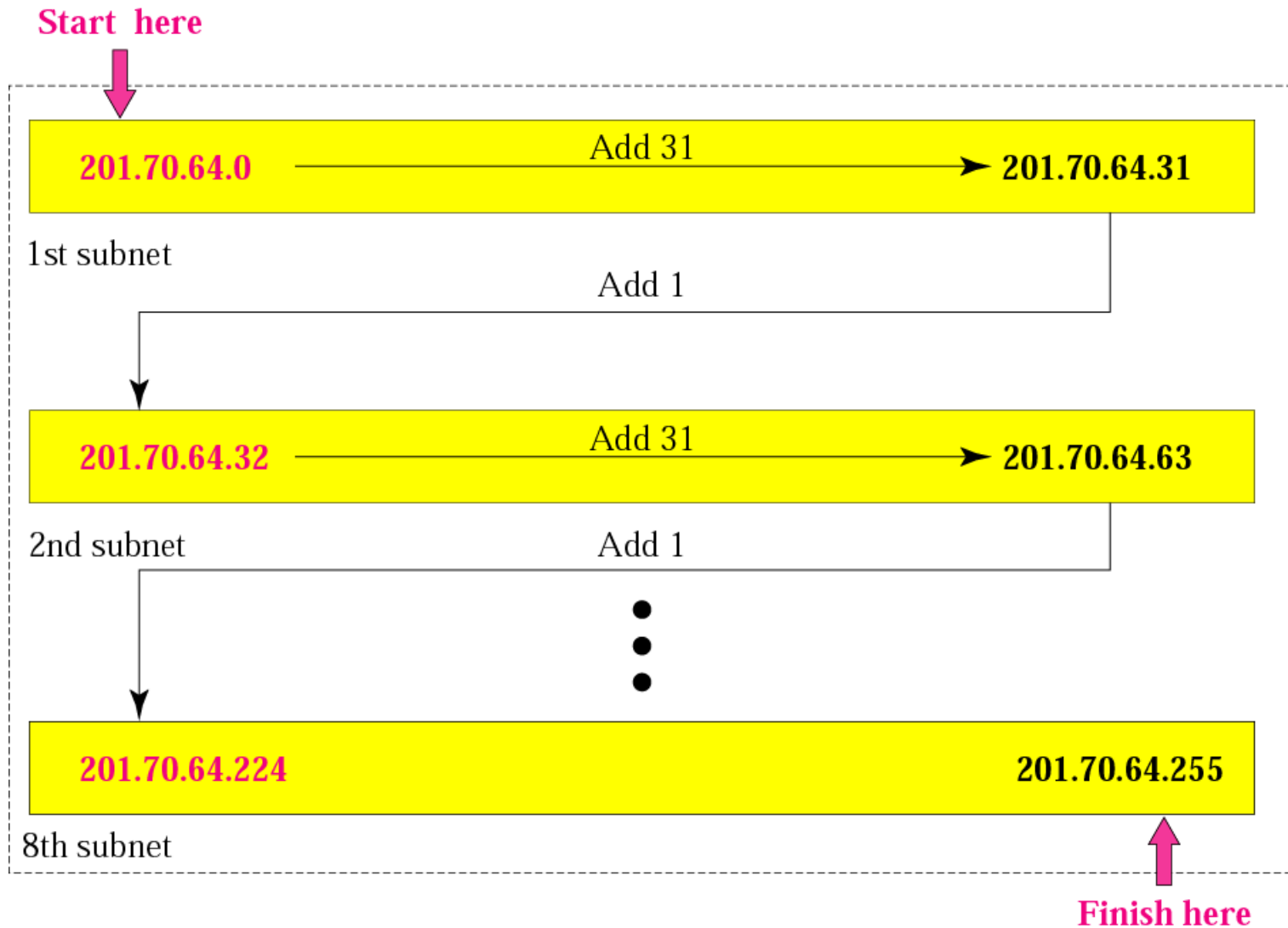
255.255.255.224

The number of subnets is 8.

The number of addresses in each subnet is 2^5 (5 is the number of 0s) or 32.

See Figure 5.8

Example 3



Example 4

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Solution

The number of 1s in the default mask is 16 (class B).

Solution (Continued)

The company needs 1000 subnets. This number is not a power of 2. The next number that is a power of 2 is 1024 (2^{10}). We need 10 more 1s in the subnet mask.

The total number of 1s in the subnet mask is 26 ($16 + 10$).

The total number of 0s is 6 ($32 - 26$).

Solution (Continued)

The mask is

11111111 11111111 11111111 11000000

or

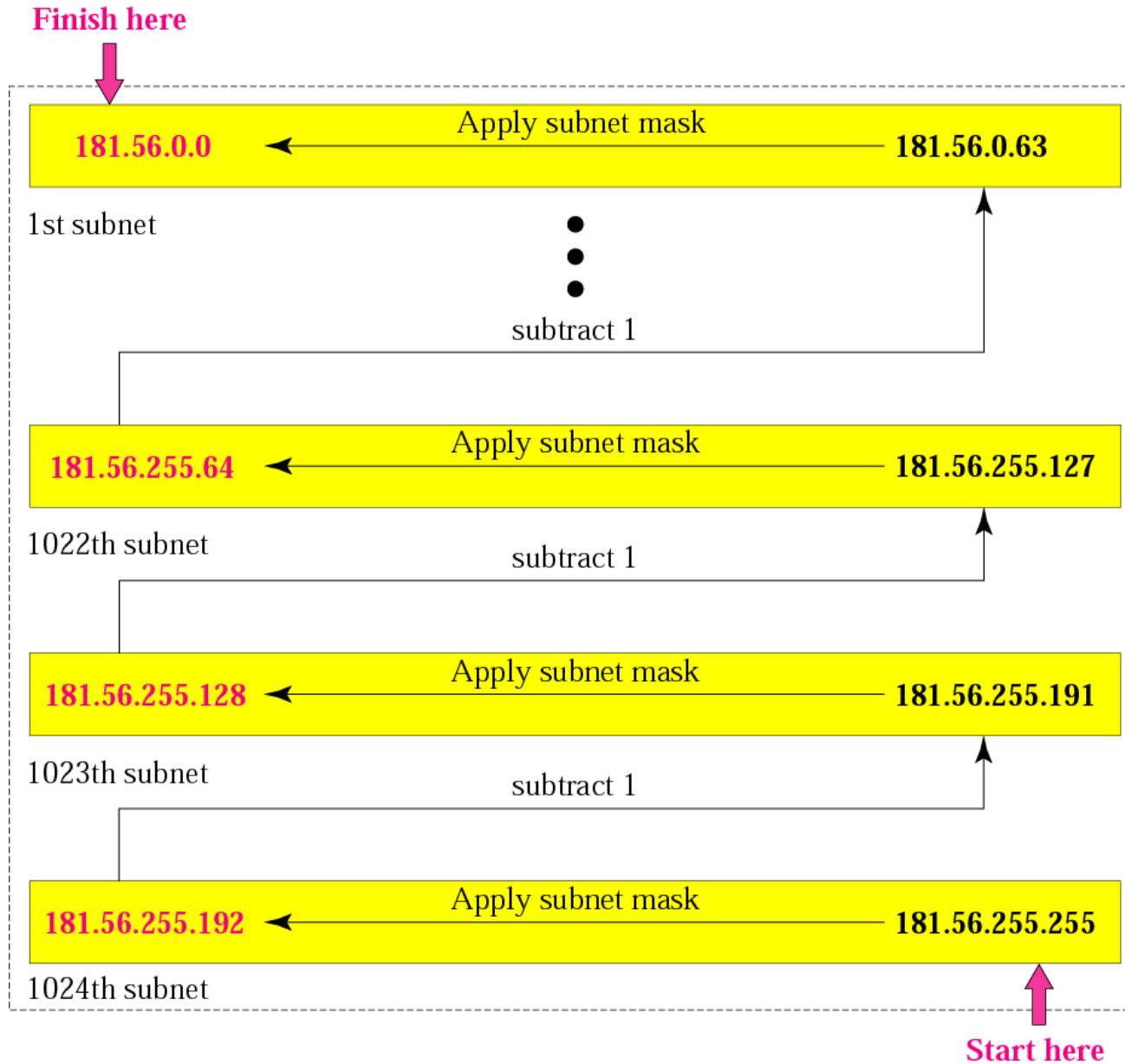
255.255.255.192.

The number of subnets is 1024.

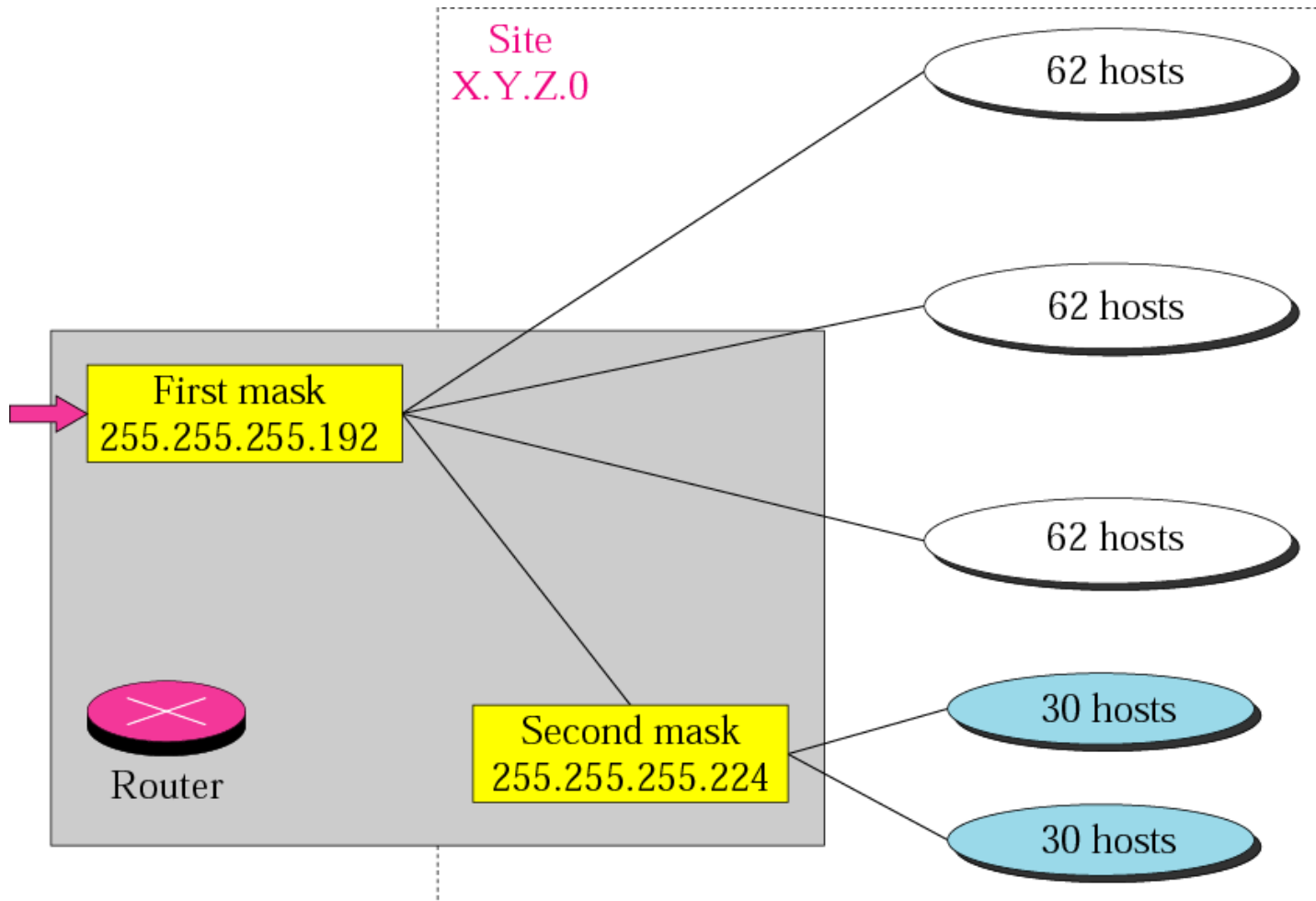
The number of addresses in each subnet is 2^6 (6 is the number of 0s) or 64.

See Figure 5.9

Example 4



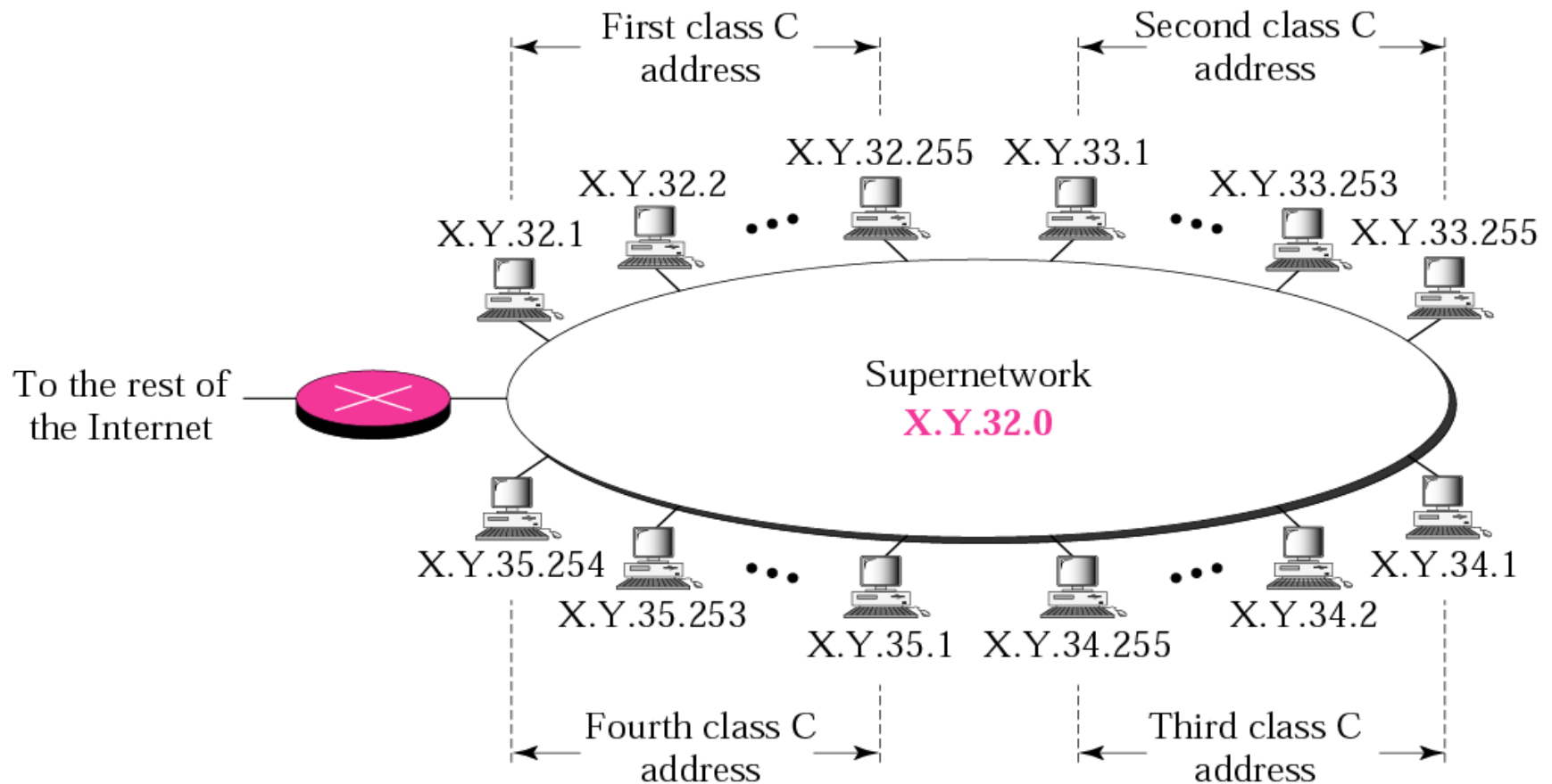
Variable-length subnetting



Note

*In subnetting,
we need the first address of the
subnet and the subnet mask to
define the range of addresses.*

Figure 4.26 *A supernetwork*



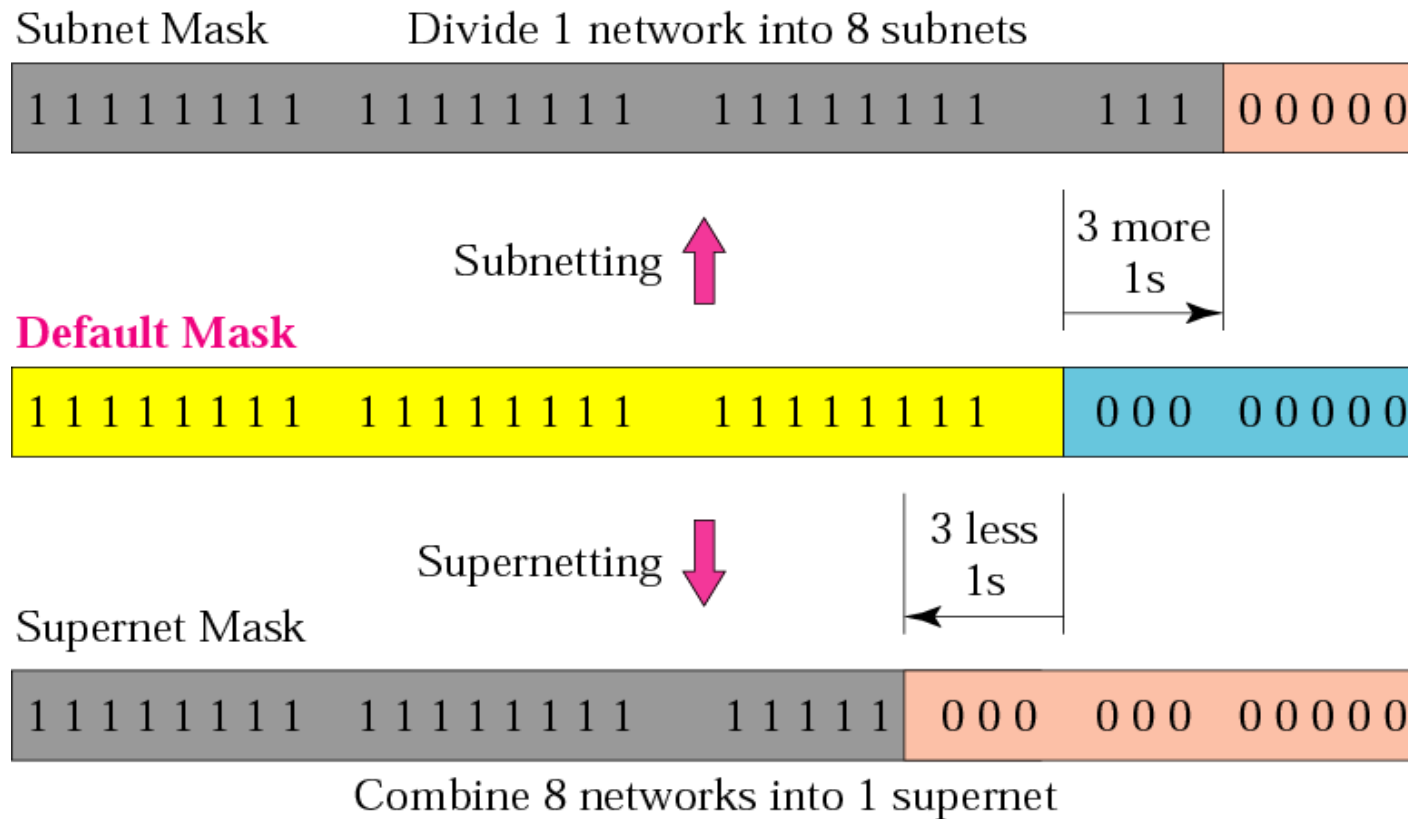


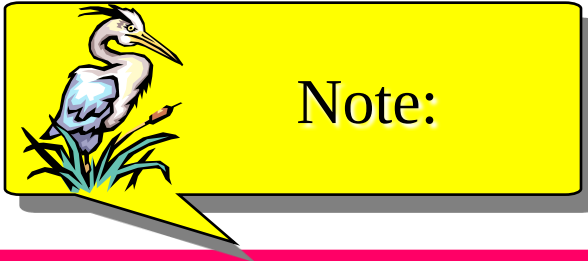
Note:

In subnetting, we need the first address of the subnet and the subnet mask to define the range of addresses.

In supernetting, we need the first address of the supernet and the supernet mask to define the range of addresses.

Figure 4.27 *Comparison of subnet, default, and supernet masks*





Note:

The idea of subnetting and supernetting of classful addresses is almost obsolete.



Note

Classful addressing, which is almost obsolete, is replaced with classless addressing.

Classless Addressing

- To overcome address depletion and give more organizations access to internet classless addressing were designed and implemented.
- There are no classes but the addresses are granted in blocks.

Address Block: An entity is granted a block(range) of addresses.

Restriction:

1. Address in block must be contiguous, one after another.
2. Number of addresses in a block must be in power of 2.
3. The first address must be evenly divisible by the number of addresses.



Example 19.5

Figure 19.3 shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.

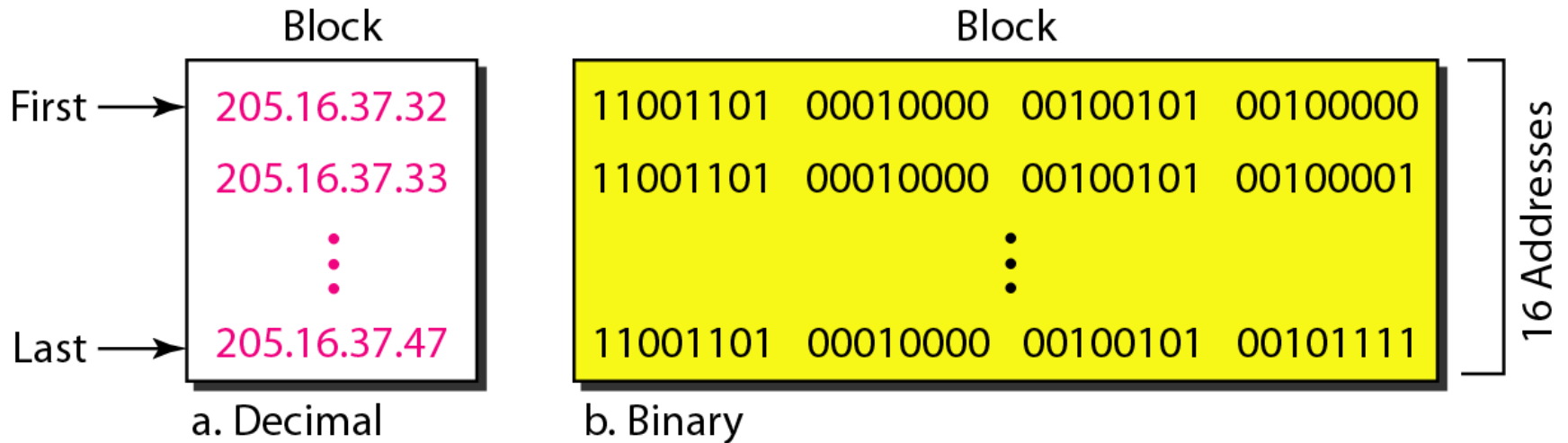
Restrictions: We can see that the restrictions are applied to this block.

The addresses are contiguous.

The number of addresses is a power of 2 ($16 = 2^4$).

The first address is divisible by 16. The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Figure 19.3 *A block of 16 addresses granted to a small organization*



Mask

- A mask is a 32 bit number in which n leftmost bits are 1s and the $32-n$ rightmost bits are 0s.
- In classless addressing the mask for a block can take any value from 0 to 32.

In IPv4 addressing, a block of addresses can be defined as

$x.y.z.t /n$

in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.

- The address and the $/n$ notation completely defines the whole block(the first address, the last address and the number of addresses).



The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.

The number of addresses in the block can be found by using the formula 2^{32-n} .



Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.



Example 19.7

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

This is actually the block shown in Figure 19.3.



Example 19.8

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.



Example 19.9

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Example 19.5 the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a.** *The first address*
- b.** *The last address*
- c.** *The number of addresses.*

Example 19.9 (continued)

Solution

- a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.*

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Example 19.9 (continued)

- b.** *The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.*

Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

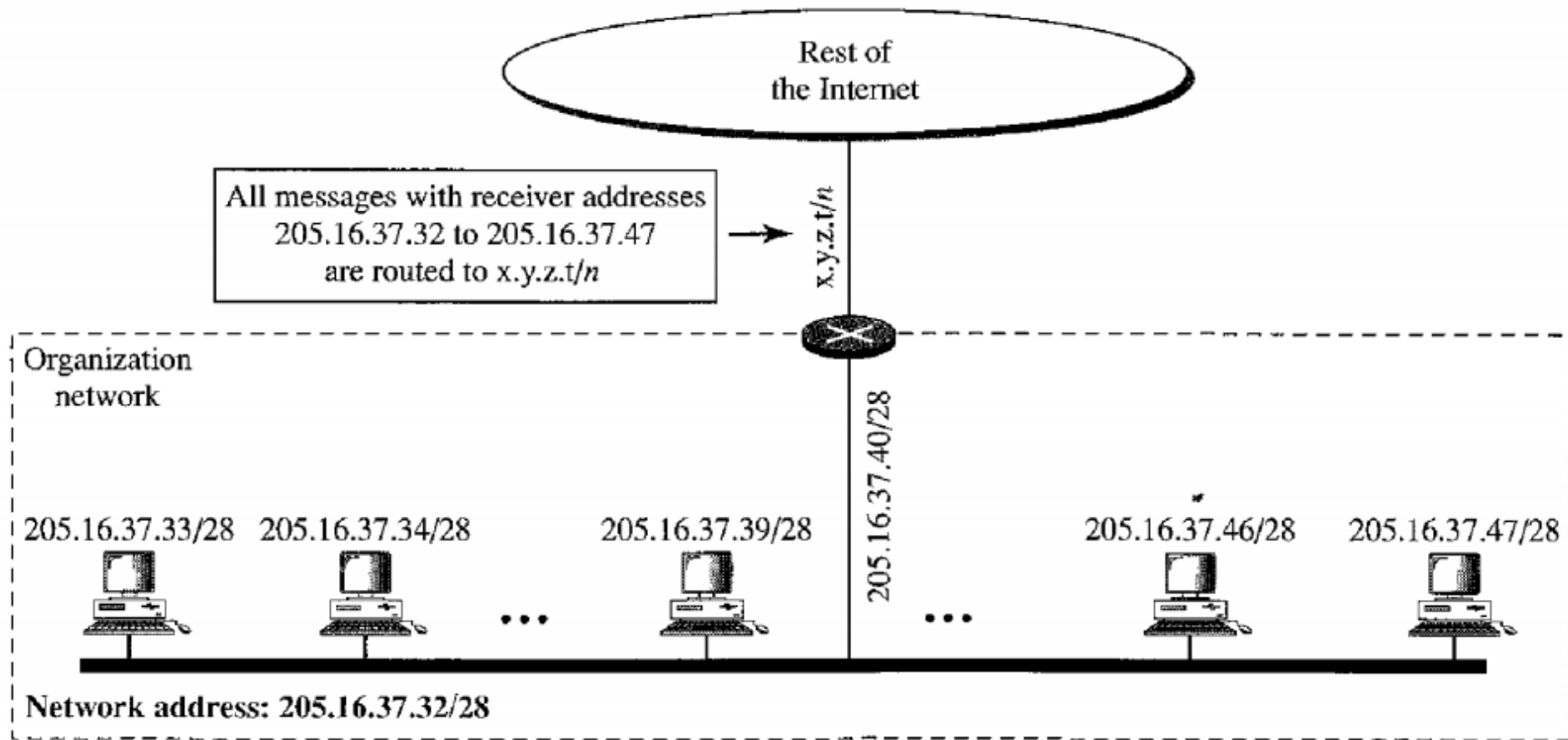


Example 19.9 (continued)

- c.** *The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.*

Mask complement:	00000000 00000000 00000000 00001111
Number of addresses:	$15 + 1 = 16$

Figure 19.4 *A network configuration for the block 205.16.37.32/28*



Note

The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

Figure 19.5 *Two levels of hierarchy in an IPv4 address*

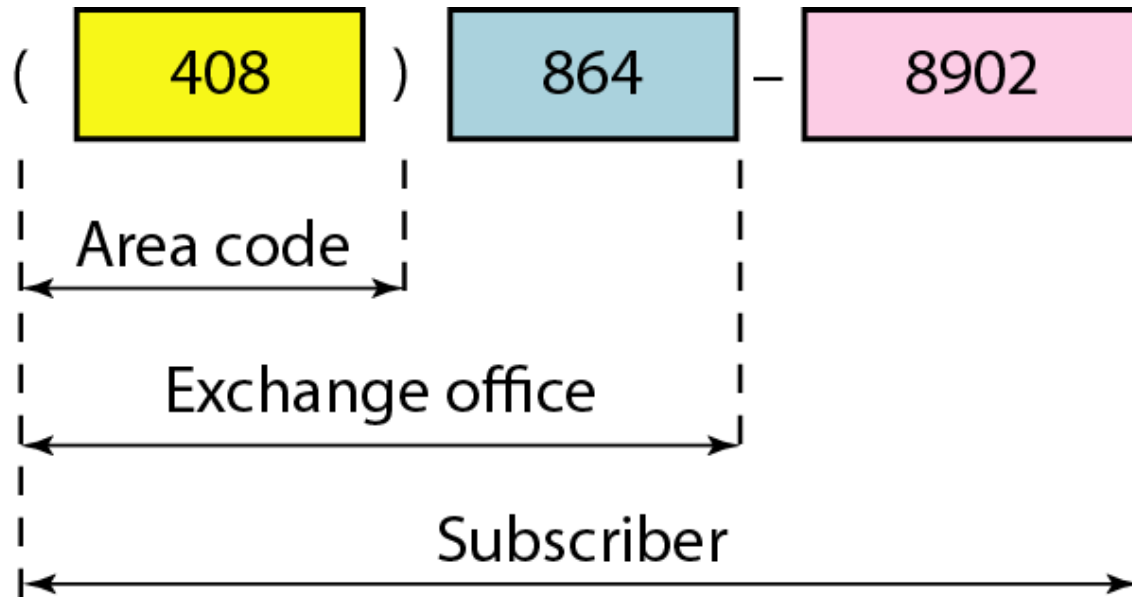
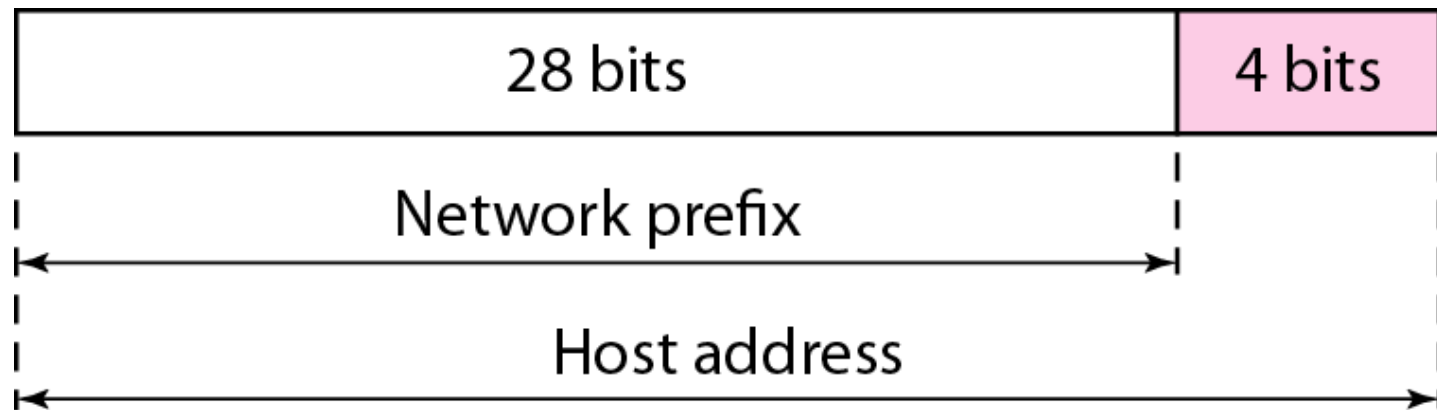


Figure 19.6 *A frame in a character-oriented protocol*



Note

**Each address in the block can be considered as a two-level hierarchical structure:
the leftmost n bits (prefix) define the network;
the rightmost $32 - n$ bits define the host.**

Figure 19.7 *Configuration and addresses in a subnetted network*

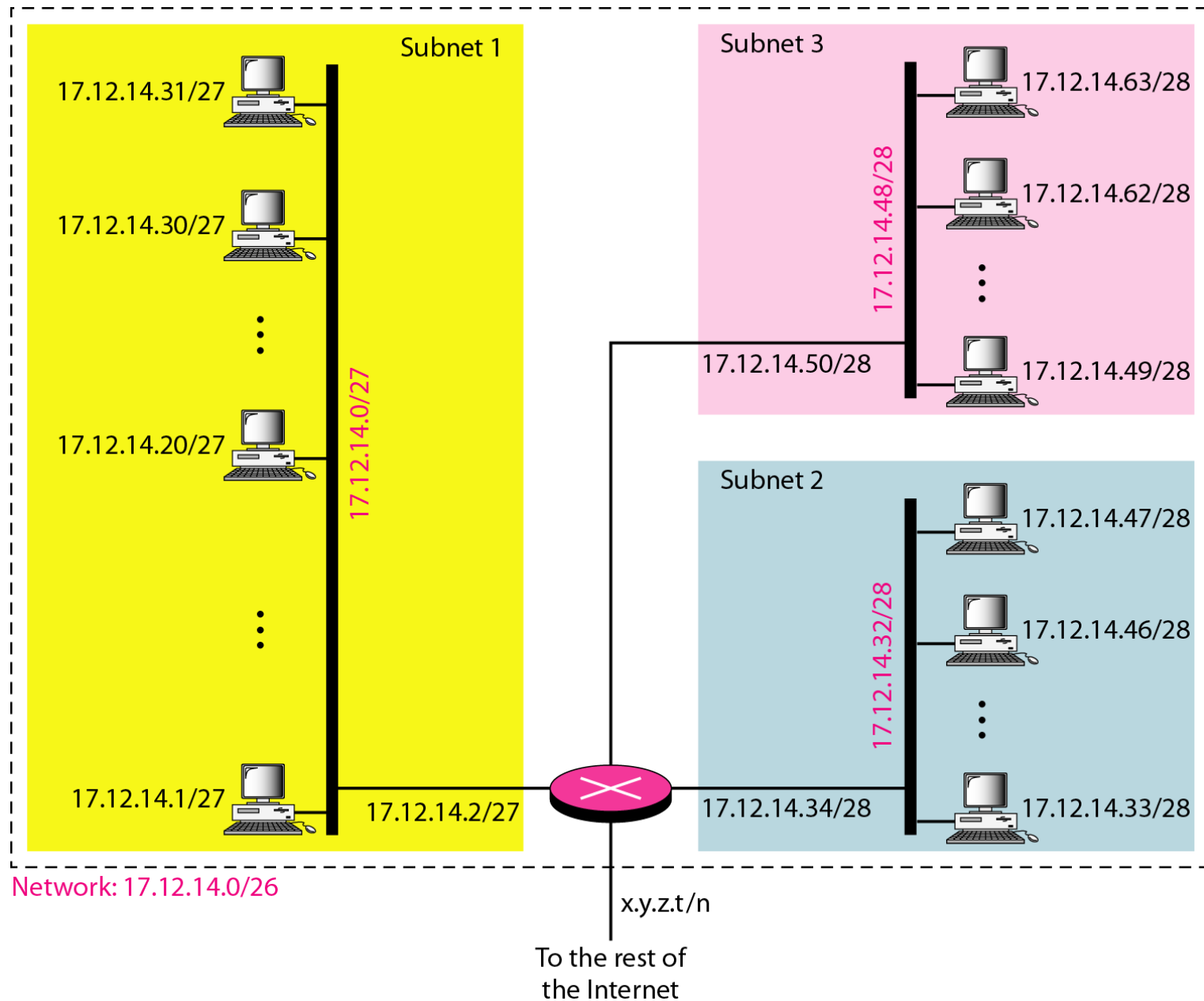
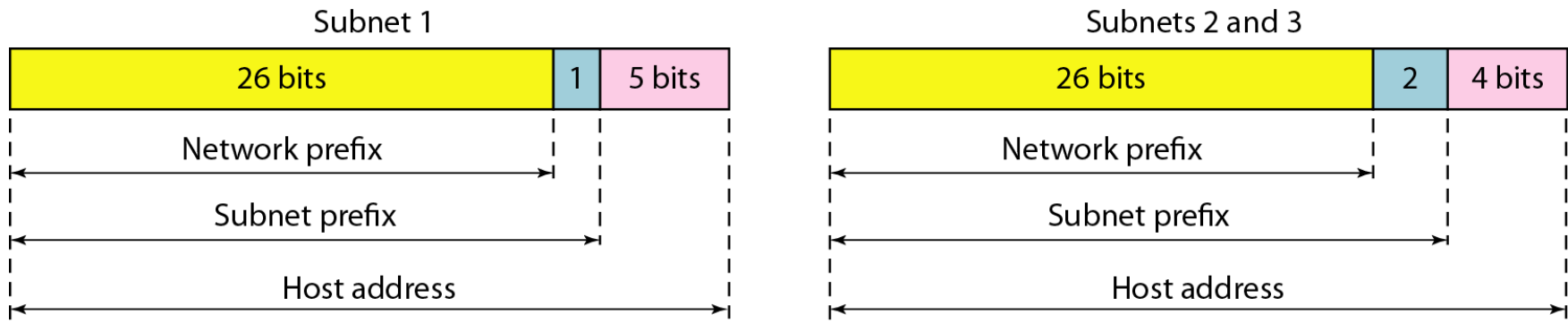


Figure 19.8 *Three-level hierarchy in an IPv4 address*





Example 19.10

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.*
- b. The second group has 128 customers; each needs 128 addresses.*
- c. The third group has 128 customers; each needs 64 addresses.*

Design the subblocks and find out how many addresses are still available after these allocations.



Example 19.10 (continued)

Solution

Figure 19.9 shows the situation.

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		



Example 19.10 (continued)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example 19.10 (continued)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

Number of available addresses: 24,576

Figure 19.9 *An example of address allocation and distribution by an ISP*

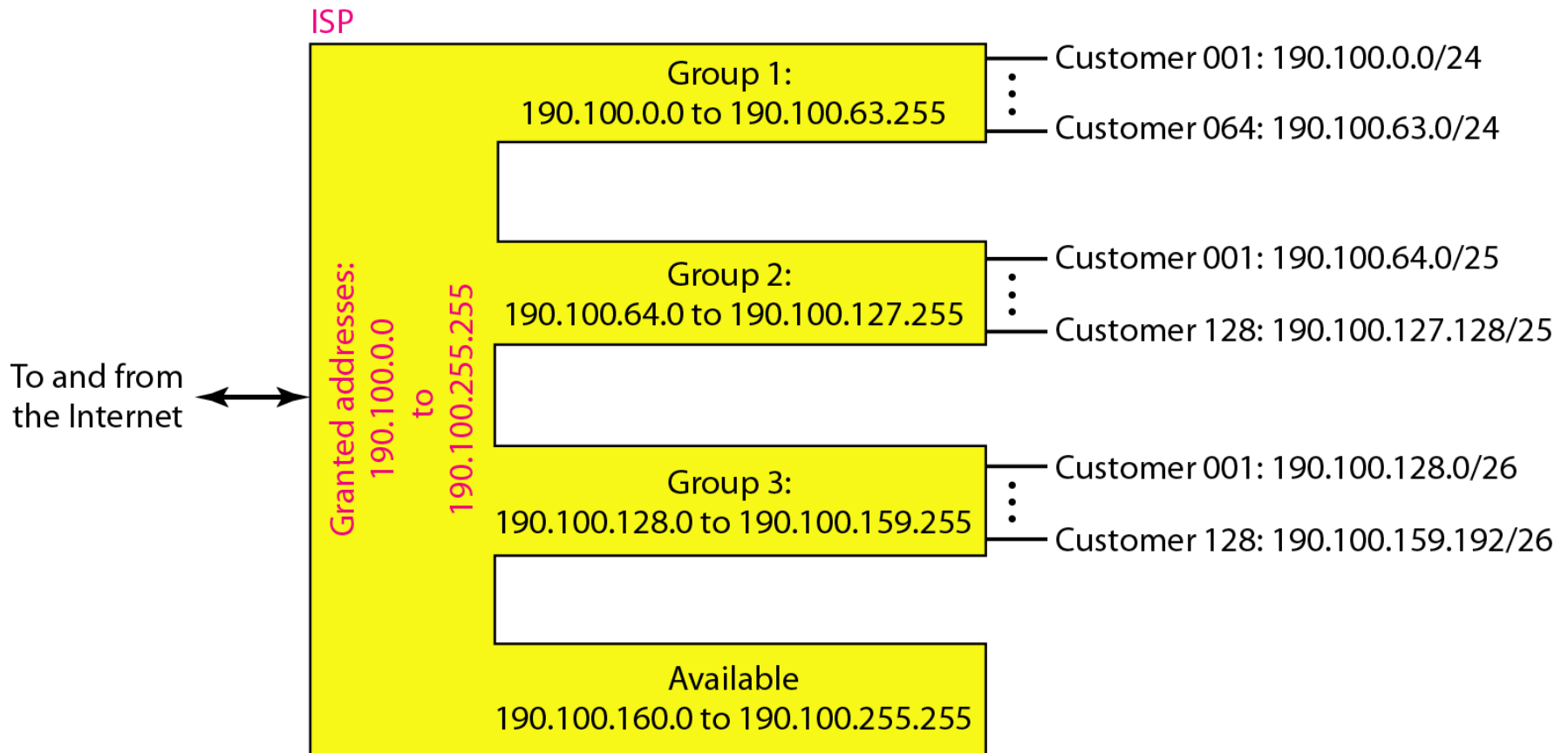


Table 19.3 *Addresses for private networks*

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

19-2 IPv6 ADDRESSES

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.

Topics discussed in this section:

Structure

Address Space



Note

An IPv6 address is 128 bits long.

Figure 19.14 *IPv6 address in binary and hexadecimal colon notation*

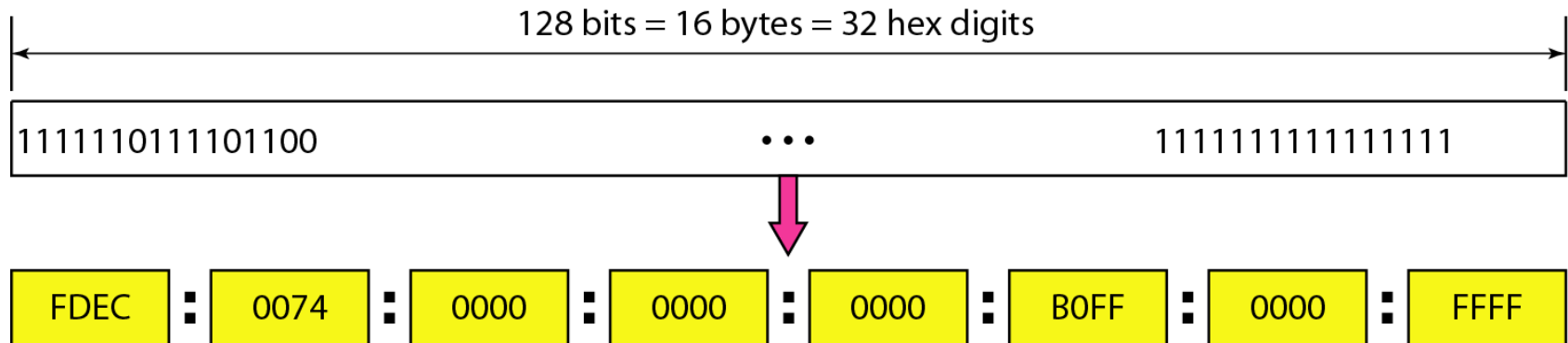
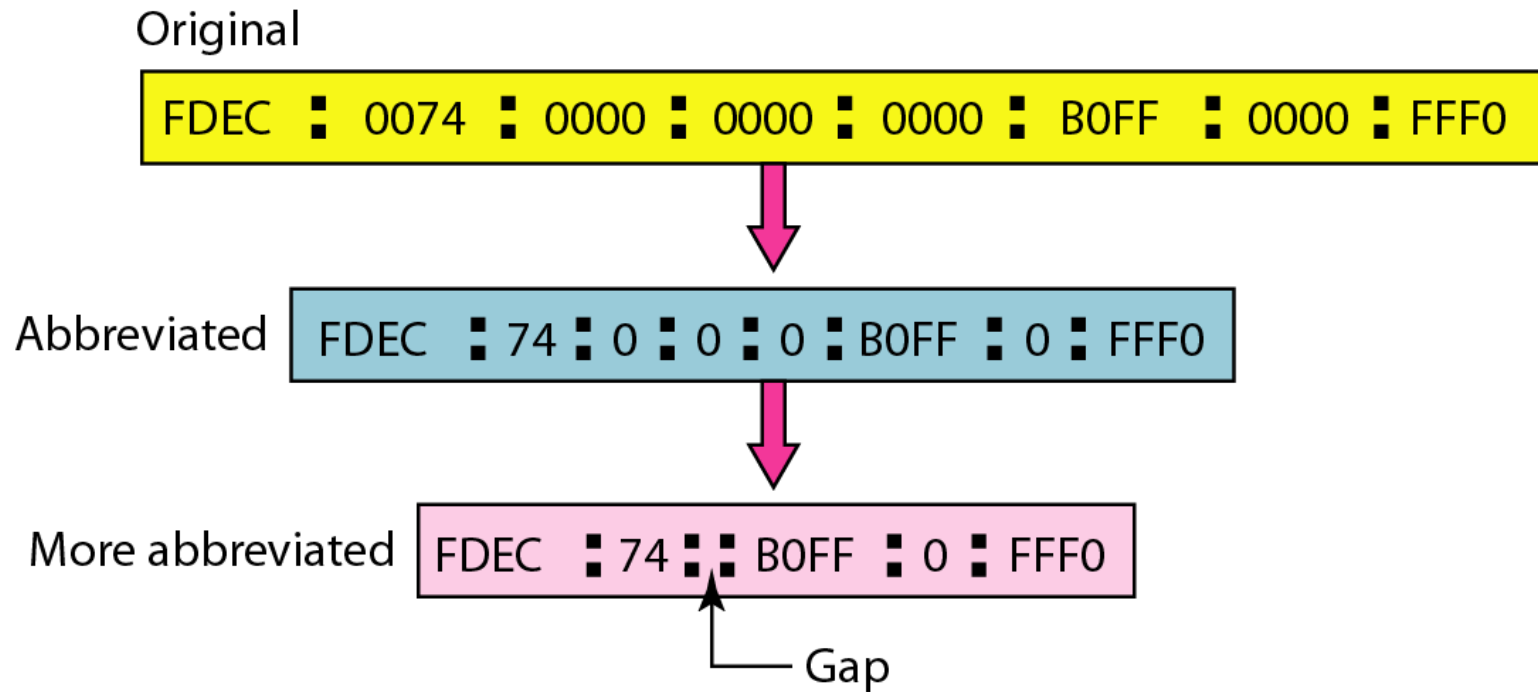


Figure 19.15 *Abbreviated IPv6 addresses*



Example 19.11

Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX
0: 15: : 1: 12:1213

This means that the original address is.

0000:0015:0000:0000:0000:0001:0012:1213
